CURRICULUM VITAE

October 24, 2026

David LACOSTE

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EDUCATION

2009	Habilitation degree from P. et M. Curie University, Paris
1996 – 1999	Ph.D. in Theoretical Physics, Université Joseph Fourier, Grenoble
1994 – 1995	Master in Quantum Physics from Ecole Normale Supérieure, Paris
1992 – 1995	Undergraduate studies at Ecole Normale Supérieure, Paris

EMPLOYMENT

Jun 2016 – present	CNRS Directeur de Recherches (DR2 level) at Gulliver laboratory, ESPCI
Jun 2005 – Jun 2016	CNRS Chargé de Recherches (CR1 level) at Gulliver laboratory, ESPCI
Jan 2002 – Jun 2005	CNRS Chargé de Recherches (CR2 level) at Gulliver laboratory, ESPCI
Jan 2000 – Jan 2002	Postdoctoral Fellow with T. Lubensky at Dept. of Physics, University of Pennsylvania in Philadelphia, working on various Soft Matter problems

HONORS

Sept 2020	Distinction "Outstanding referee" for APS
Sept 2012	Distinction "Outstanding referee" for EPL
2011 – 2015	Distinction "Prime d'excellence scientifique" from CNRS

RESEARCH

I am a theoretical physicist working on the statistical physics of living matter and on the emergence of life. My research spans stochastic thermodynamics, the origin of life, chemical reaction networks and autocatalysis, and the growth and fluctuations of cell populations. The line of work on autocatalysis also led to an on-going study of the limits to growth in economics. I am the author of 88 peer-reviewed publications (h-index 30), available on my web site `https://dlacoste22.github.io/publications.html` and on Google Scholar.

SELECTED PUBLICATIONS

[1] A. Blokhuis, D. Lacoste and P. Nghe, *Universal motifs and the diversity of autocatalytic systems*, Proc. Natl. Acad. Sci. U.S.A. **117**, 25230–25236 (2020).

[2] L.-F. Rojas-Ochoa, D. Lacoste, R. Lenke, P. Schurtenberger and F. Scheffold, *Depolarization of backscattered linearly polarized light*, J. Opt. Soc. Am. A **21**, 1799–1804 (2004).

[3] F. Quemeneur, J. K. Sigurdsson, M. Renner, P. J. Atzberger, P. Bassereau and D. Lacoste, *Shape matters in protein mobility within membranes*, Proc. Natl. Acad. Sci. U.S.A. **111**, 5083–5087 (2014).

[4] A. W. C. Lau, D. Lacoste and K. Mallick, *Nonequilibrium fluctuations and mechanochemical couplings of a molecular motor*, Phys. Rev. Lett. **99**, 158102 (2007).

[5] M. F. Islam, K. H. Lin, D. Lacoste, T. C. Lubensky and A. G. Yodh, *Field-induced structures in miscible ferrofluid suspensions with and without latex spheres*, Phys. Rev. E **67**, 021402 (2003).

ONGOING GRANTS

Sept 2026 – Aug 2029	Coordinator of the MERCASTO project (PSL University), *Network-based detection of economic sub-structures responsible for carbon emissions*, in collaboration with S. Benoit (Université Paris Dauphine - PSL); funds the PhD of Jiwei Wang.
Oct 2025 – Sept 2029	Coordinator of the ANR project *Porous life*, on the emergence of life from confined autocatalytic chemical reactions.

TEACHING IN THE LAST FIVE YEARS

2025 – ongoing	Tutorials on statistics, and lectures on energy transition for 1st year ESPCI students and master students
2016 – ongoing	Biophysics course for 3rd year students in ESPCI
2023 – 2024	Tutorials and lectures on Energy conversion in Paris-Saclay university
2021 – 2023	Organization and lectures for PSL weeks in ESPCI
2021 – 2022	Tutorials on Computation and Data Driven Biophysics at ENS

STUDENT SUPERVISION

I currently supervise 2 PhD students: B. Ledoux, who is in his last year, and Jiwei Wang, who started in September 2026. I have supervised 6 PhD students and about 9 postdocs so far.

INVITED TALKS SINCE 2022

- Thermodynamics 2.0, Boone, NC, USA, August 2026
- IFOM workshop: From Cells to Populations: Quantitative Principles of Growth and Evolution, Milan, January 2026
- Conference on Extremal Statistics in Biology, Vienna, June 2025
- Workshop on Action vs. Inaction facing climate change, Zinal, Switzerland, May 2025
- Conference on biological information processing, Okinawa, Japan, April 2025
- Solvay Workshop in honor of P. Gaspard, Brussels, December 2024
- Evolution in Paris, Paris, September 2024
- Paris-Rice meeting, Paris, September 2024
- Thermodynamics 2.0 (online), Broomfield, USA, August 2024
- Workshop on Emergence of Information in Molecular Systems, MIAPP program, Garching, July 2024
- Workshop on Evolution of population dynamics, Pasteur Institute, Paris, June 2024
- Congress of the French Physical Society, Paris, July 2023
- Postmodern Thermodynamics conference, Luxembourg, December 2022
- Thermodynamics 2.0 (online), Riverside, USA, July 2022
- Stochastic Thermodynamics conference (online), ICTP, Bangalore, June 2022

ACTIVITIES OF GENERAL INTEREST

- Member of two scientific committees of Paris Sciences et Lettres University (PSL), Qlife and Smart-waves, which concern biophysics and the physics of waves respectively.
- Member of committee "Objectif décarbonation" at ESPCI.
- Organization of conferences: 3 in 2017-2018; 2 in 2024, including a celebration of "Sadi Carnot's legacy" at Ecole Polytechnique, September 16-18, 2024; and 2 in 2026: a summer school on Pattern formation in biology in Beg Rohu, Brittany, in May 2026, and a conference on "Thermodynamics and Kinetics of Chemical Reaction Networks: from Cell Metabolism to Design Principles for Synthetic Life" in Garching in September 2026.